H2GO Case Study

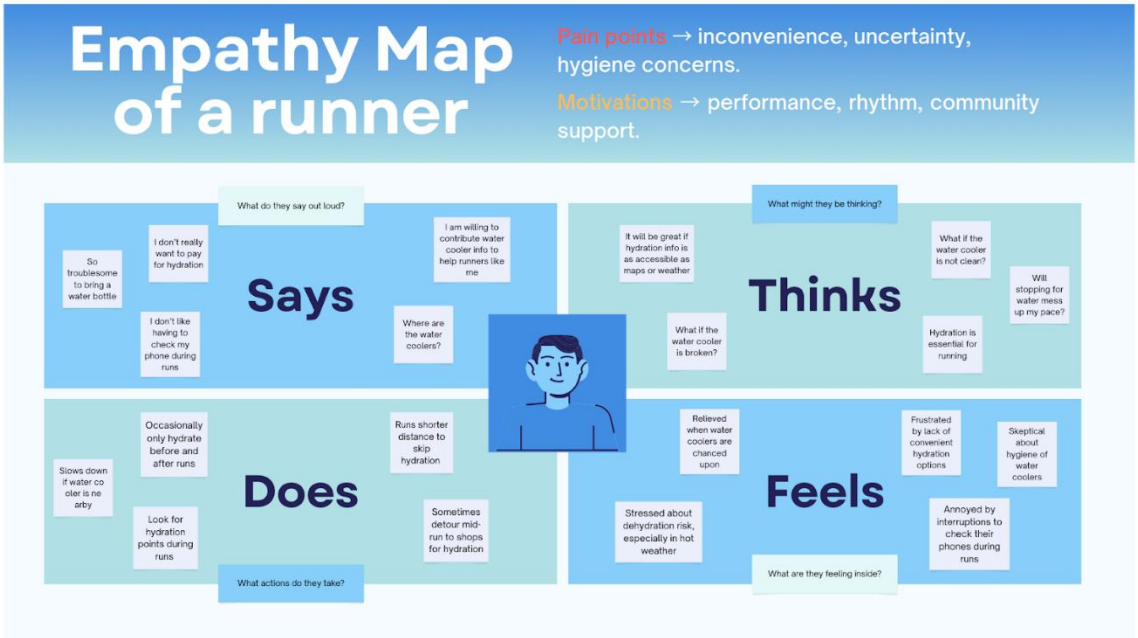
PROJECT TITLE:

Problem : In Singapore's hot and humid climate, runners often struggle to stay hydrated without carrying heavy water bottles that disrupt their pace and comfort. According to the Half-Marathon and Full-Marathon Runners' Hydration Practices and Perceptions study (O'Neal et al., 2011, PubMed Central), 70% of runners reported major performance drops due to dehydration, and 45% experienced dehydration-related health symptoms during runs. This highlights a lack of hydration awareness and accessibility amongst endurance runners. This finding emphasizes the need for runners to have a convenient, hygienic and accessible method to find hydration points along their running routes.

Solution: An app that integrates water cooler location into navigation and customizable route planning, making hydration stops convenient and minimally disruptive. It also features saved routes, hygiene ratings, functionality reviews and rerouting during run.

USER RESEARCH AND EMPATHY:

To start this project, we embarked on some user research and noted the pain points and motivations of our target groups. With this, we devised the empathy map:



Subsequently, we conjured up a persona(Ryan Koh), allowing us to guide our design decisions and solution centered around a singular profile, hence designing for a real problem instead of too wide of an audience.

RYAN KOH

YEAR 1 STUDENT
AT SMU

A recreational runner who usually runs at an easy pace where he can still have simple conversations while running

Runs 3 times a week
(each session at least 8km)

PAIN POINTS

- Finds it troublesome to bring a water bottle on his run
- Has difficulty finding hydration points during run, risk cutting short run due to dehydration
- Avoids using water coolers that are visibly unclean and poorly-maintained

GOALS AND NEEDS

- Maintain overall health and well-being
- At least 2 hydration points along running route to maintain hydration
- Build endurance and stamina
- Tracking of personal fitness progress

INTERESTS

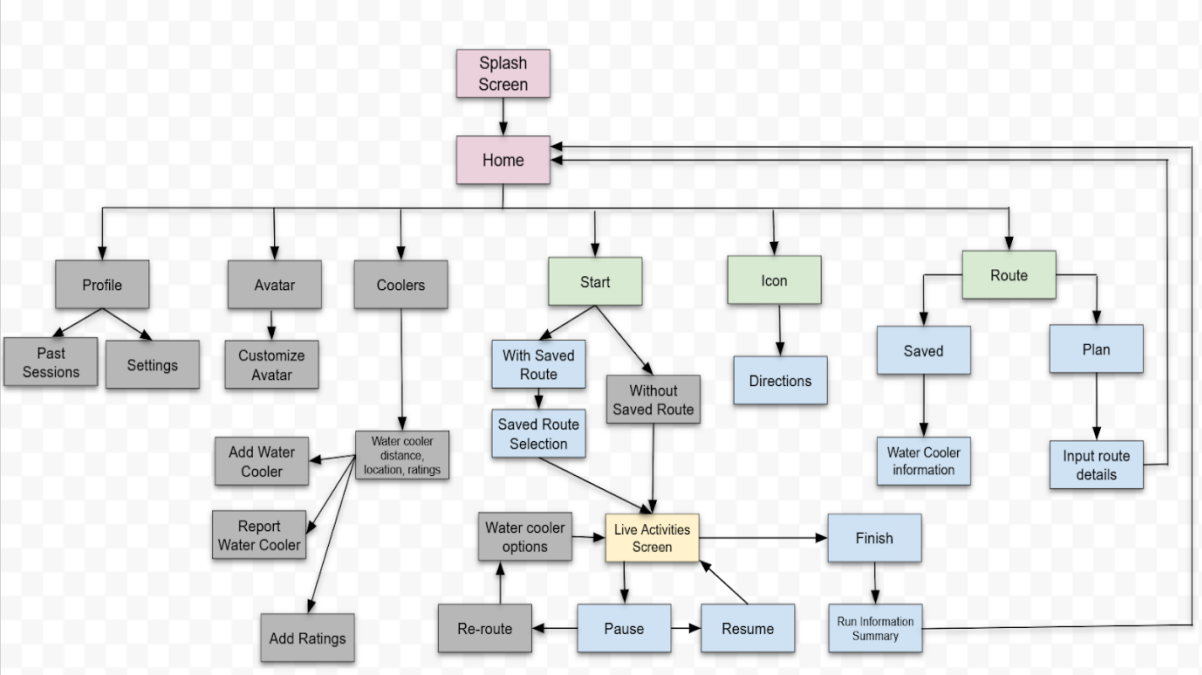
- Distance running
- Explore new running routes

RELEVANT TECHNOLOGY USED

- Strava for running route and progress tracking

INFORMATION ARCHITECTURE:

Next, we started designing the solution and made sure to map out a navigation diagram so that the group is clear on the requirements and all the features we are building. It also allows us to understand and simplify how our target user would move through this app.



PROTOTYPING AND ITERATIVE DESIGN:

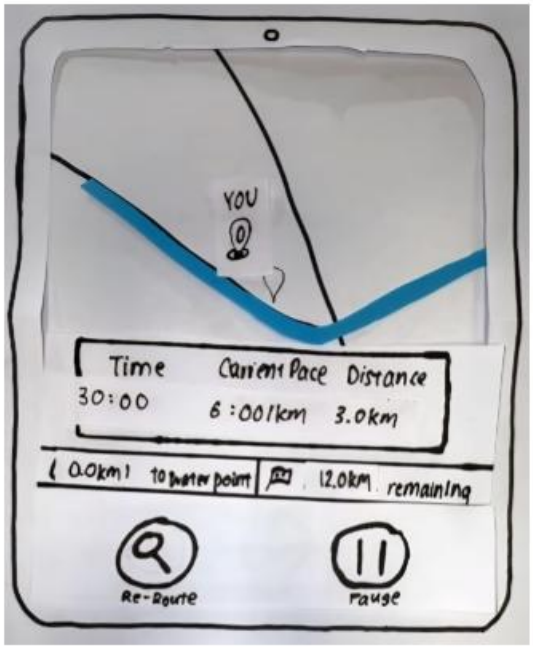
Subsequently, we did some lo-fi prototyping to rapidly develop the most viable solution for us to build. We also made use of Heuristic Evaluation to understand the feedback of users and improve our final design. We highlighted major issues and the recommendations that followed.

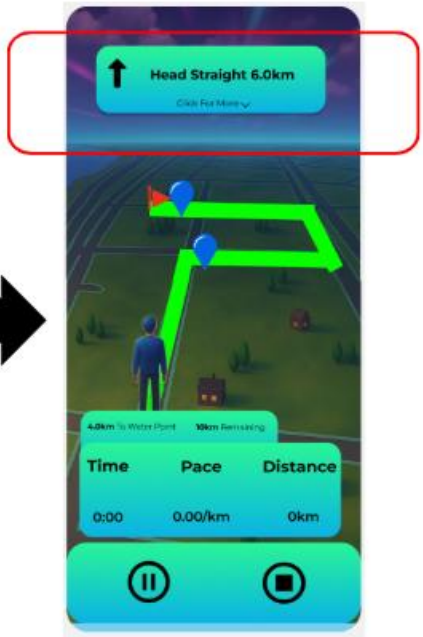
1. **Detailed Routing Instructions**

   Description: Visual navigation system that shows the exact navigation instructions to a water cooler during a run.

   Justification: Heuristic Evaluation #2. To provide users with detailed instructions so users know exactly how to reach the target water cooler.

   Prototype Version(s): Prototype 1.0.



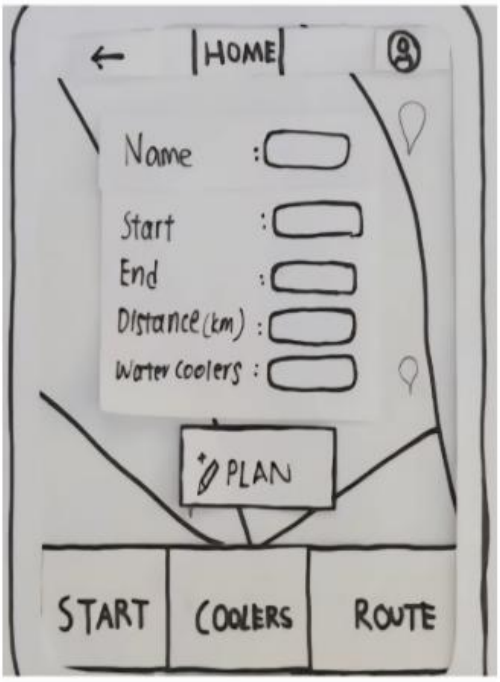


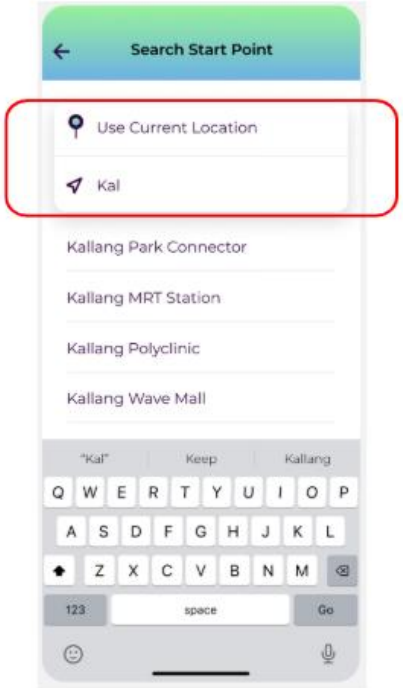
2. **Search Location Input on Map Screen**

   Description: Add a search bar for users to search for the start and end locations during the planning of a run

   Justification: Heuristic Evaluation #3. To provide an alternative to manually scrolling on the screen, and significantly improve time efficiency of map planning.

   Prototype Version(s): Prototype 1.0.



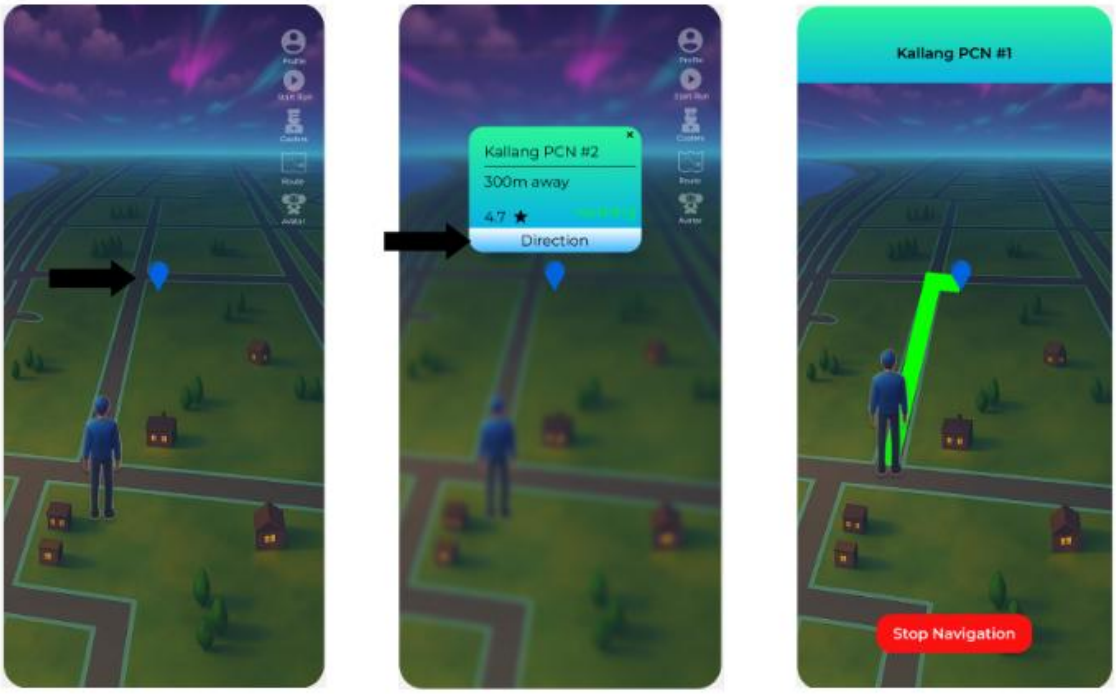


3. **Water Cooler On-Click Navigation**

Description: Clicking on a nearby water cooler will immediately highlight a path and provide detailed routing instructions to it.

Justification: To provide an efficient and quick method for users to navigate to water coolers.

Prototype Version(s): Prototype 1.0.



Subsequently, we conducted usability testing using Loop11 to test some design choices and collected data to guide our decision. We conducted A/B testing with 40 participants and stuck with version A for visual simplicity.

Usability Testing / Experiment:

The goal of this experiment is to determine whether a water droplet icon or a water cooler icon will allow users to complete the task of navigating to a water cooler more effectively and easily.

Variables

Independent Variable

Our independent variable is the icons used to represent water coolers available on the map. We aim to understand how these different visual designs affect the efficiency of users finding a water cooler.

- **Version A (Control):** This is the current prototype format, the icon for water coolers is represented by a water droplet.
- **Version B (Alternative):** A new prototype format where the icon for water coolers is represented by a miniature water cooler instead.

Experiment Screenshots

Version A (Water Droplet Icon)	Version B (Water Cooler Icon)

The Result: A clickable prototype that simplifies route planning with built-in hydration stops.

FIGMA PROTOTYPE: